Supplementary Material: Single-Channel EEG Detection of N3 Sleep with Interpretable Slow-Wave Factor Scores

This supplement follows the order of the Results in the main text; the tag at the start of each caption gives the research question it supports. All values come from run `20261030-1528`, and the complete tables are in the repository folder `v2/results/20261030-1528`. Analyses marked post hoc were added after the first complete run and do not enter any prespecified verdict.

Topic (main text)	Figures	Tables
Baselines (III-D, IV-B)	–	SI
Detection (IV-A, IV-B)	S1 curves, S2 calibration, S3 participants	–
Factor scores (IV-C, IV-D)	S4 decline, S5 N2 vs N3	SIII grade shares
Stability and profile (IV-F, IV-J)	S6 minutes and retest, S10 example night	SIV agreement (ICC)
Components (IV-G)	S7 leave-one-out	SII effects
Explanations (IV-H)	S8 band importance	–
Reliability (IV-I)	S9 selective prediction	SV inputs
Robustness (IV-K)	–	SVI sensitivity

TABLE SI
[BASELINES (SECTION IV-B)] DEVIATIONS OF THE RE-IMPLEMENTED BASELINES FROM THEIR PUBLISHED TRAINING RECIPES. ALL SHARE THE FOLDS, INNER SELECTION, TEMPERATURE SCALING, THRESHOLD RULE, LOSS, CLASS WEIGHT AND AUGMENTATION OF THE PROPOSED MODEL.

Model	Implementation and deviations
DeepSleepNet	Dual-branch CNN (small and large filters) with two BiLSTM(512) layers and residual shortcut; trained end-to-end on 64-epoch sequences instead of the two-step schedule.
TinySleepNet	CNN (128 filters, 4 layers) with LSTM(128); 64-epoch sequences instead of 15–20; common amplitude/noise augmentation instead of signal shifts.
AttnSleep	Multi-resolution CNN, feature recalibration and causal temporal context encoder for 100 Hz; epoch-wise as published; weighted binary instead of weighted categorical cross-entropy; encoder in 32-bit precision.
L-SeqSleepNet	STFT input, learnable filterbank, BiGRU epoch encoder and long-sequence encoder; 64 epochs folded into 8 subsequences of 8 instead of 200 as 20 of 10.

TABLE SII
COMPONENT EFFECTS: MEDIAN PAIRED PARTICIPANT DIFFERENCE IN F1 (WITH MINUS WITHOUT), 95% CI AND HOLM-CORRECTED p. UPPER BLOCK PRESPECIFIED; LOWER BLOCK POST HOC, FROM THE FULL MODEL, WITH ITS OWN HOLM FAMILY.

Component	ΔF1 (95% CI)	p
Feature fusion (no context)	+0.003 (−0.001, 0.006)	1
Class weighting (no context)	0.000 (−0.003, 0.003)	1
Attention vs mean pooling	−0.003 (−0.006, 0.000)	0.34
Temporal context	+0.023 (0.018, 0.027)	<0.001
CNN vs features-only boosting	+0.036 (0.030, 0.043)	<0.001
Feature fusion (full model)	+0.007 (0.003, 0.011)	0.018
Class weighting (full model)	0.000 (−0.003, 0.004)	1
Attention (full model)	+0.003 (0.000, 0.006)	1
Context ±15 vs ±3 epochs	+0.003 (0.001, 0.005)	1
All 19 vs 10 delta features	+0.002 (−0.002, 0.004)	1

TABLE SIII
SLOW-WAVE GRADE DISTRIBUTION OF DETECTED N3 EPOCHS (SHARE OF EPOCHS).

Cohort	Low	Moderate	High
HMC (CV)	0.529	0.377	0.094
ISRUC-I (CV)	0.653	0.285	0.062
CAP	0.523	0.353	0.124
DOD-H	0.560	0.359	0.081
DOD-O	0.746	0.215	0.039
ISRUC-II	0.625	0.289	0.086
ISRUC-III	1.000	0.000	0.000

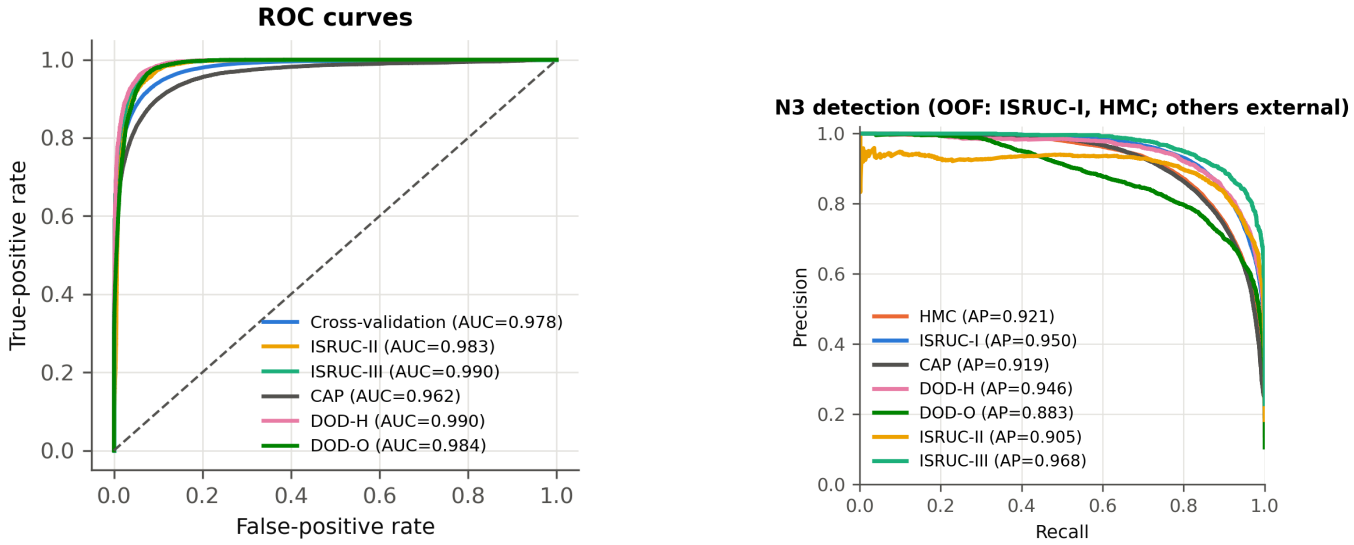


Fig. S1. ROC curves (left) and precision–recall curves (right) per cohort.

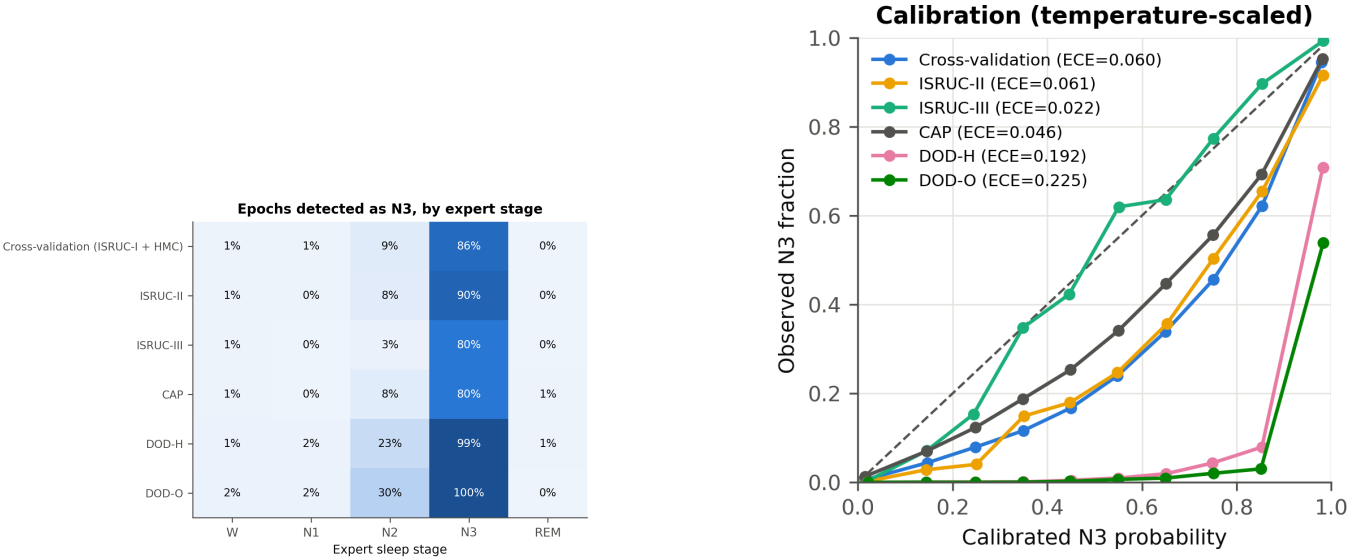


Fig. S2. Left: share of epochs of each expert stage predicted as N3. Right: reliability diagram of the temperature-scaled N3 probability with expected calibration error per cohort.

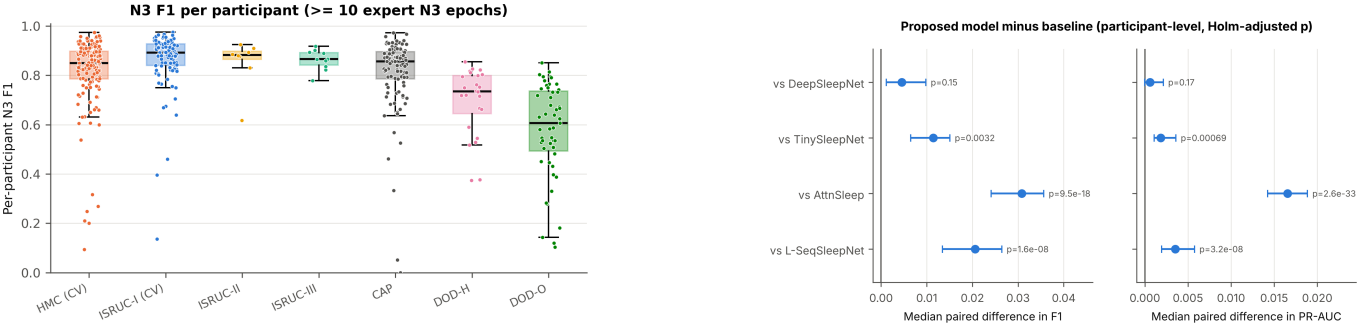


Fig. S3. Left: participant-level F1 scores (participants with at least 10 expert N3 epochs). Right: paired participant-level differences between the proposed model and each baseline with 95% CI and Holm-corrected p.

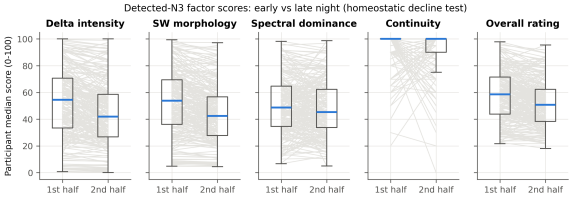


Fig. S4. Participant median factor scores of detected N3 in the first and second half of the night (cross-validation).

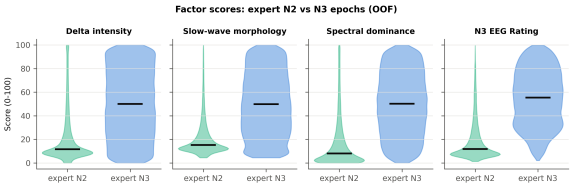


Fig. S5. Factor scores of expert N2 and expert N3 epochs (cross-validation).

TABLE SIV
INTRACLASS CORRELATION ICC(2,1) WITH MCGRAW–WONG 95% CONFIDENCE INTERVALS FOR OVERNIGHT-PROFILE AGREEMENT (PREDICTED VS EXPERT HYPNOGRAM) AND ISRUC-II TEST–RETEST.

Set	Quantity	ICC	95% CI	n
CV	N3 minutes	0.806	0.76 to 0.85	251
CV	Longest episode	0.780	0.72 to 0.83	251
CV	Episodes	0.622	0.53 to 0.70	251
CV	Median rating	0.950	0.94 to 0.96	249
CAP	N3 minutes	0.730	0.63 to 0.81	105
CAP	Longest episode	0.765	0.67 to 0.83	105
CAP	Episodes	0.700	0.59 to 0.79	105
CAP	Median rating	0.888	0.83 to 0.93	105
DOD-H	N3 minutes	0.403	-0.07 to 0.77	25
DOD-H	Longest episode	0.542	-0.04 to 0.81	25
DOD-H	Episodes	0.381	-0.10 to 0.71	25
DOD-H	Median rating	0.721	-0.07 to 0.93	24
DOD-O	N3 minutes	0.222	-0.08 to 0.55	55
DOD-O	Longest episode	0.294	-0.10 to 0.61	55
DOD-O	Episodes	0.323	-0.03 to 0.58	55
DOD-O	Median rating	0.745	0.27 to 0.89	53
ISRUC-II night 1	N3 minutes	0.898	0.57 to 0.98	8
ISRUC-II night 2	N3 minutes	0.740	0.20 to 0.94	8
ISRUC-II night 1	Median rating	0.991	0.96 to 1.00	8
ISRUC-II night 2	Median rating	0.989	0.94 to 1.00	7
ISRUC-III	N3 minutes	0.794	0.05 to 0.95	10
ISRUC-III	Longest episode	0.922	0.72 to 0.98	10
ISRUC-III	Episodes	0.478	-0.07 to 0.83	10
ISRUC-III	Median rating	0.784	-0.00 to 0.95	10
Retest ISRUC-II	Rating	0.763	0.23 to 0.95	8
Retest ISRUC-II	Predicted N3 min	0.658	0.01 to 0.92	8
Retest ISRUC-II	Expert N3 min	-0.008	-0.63 to 0.65	8
Retest ISRUC-II	Delta intensity	0.791	0.32 to 0.95	8
Retest ISRUC-II	Morphology	0.754	0.23 to 0.94	8
Retest ISRUC-II	Spectral dominance	0.339	-0.26 to 0.80	8

TABLE SV
ERROR-DETECTION AREA UNDER THE ROC CURVE OF EACH RELIABILITY INPUT AND OF THE CROSS-FITTED COMBINED SCORE (CROSS-VALIDATION), AND OF THE COMBINED SCORE IN EXTERNAL COHORTS.

Input	Area	Set (combined)	Area
Confidence	0.872	CAP	0.853
MC-dropout entropy	0.879	ISRUC-II	0.872
Mutual information	0.800	ISRUC-III	0.941
Mahalanobis distance	0.855	DOD-H	0.751
QC score	0.480	DOD-O	0.689
QC high-frequency ratio	0.297	ISRUC-I disagreement	0.883
QC clipping	0.479	External disagreement	0.628
Any QC flag	0.447		
Combined	0.879		

TABLE SVI
[ROBUSTNESS, POST HOC] POST HOC SENSITIVITY ANALYSES: NUMBER OF SETTINGS THAT TEST A CONCLUSION AND NUMBER IN WHICH IT HOLDS.

Question	Analysis	Tested	Holds
Detection	decision threshold	8	7
Detection	reference standard	25	20
Detection	quality control	8	7
Detection	participant-level summary	8	7
Factor validity	early $>$ late-night decline	65	48
Grade	stage 4 vs 3 area	1	1
Stability	weighting, factor removal	8	7
Components	leave-one-component-out	5	1
Explanations	baselines, steps, outcomes, bootstrap	11	11
Reliability	error detection per cohort	9	8
Profile	N3 minutes ICC	16	10
Profile	episode definition	16	2

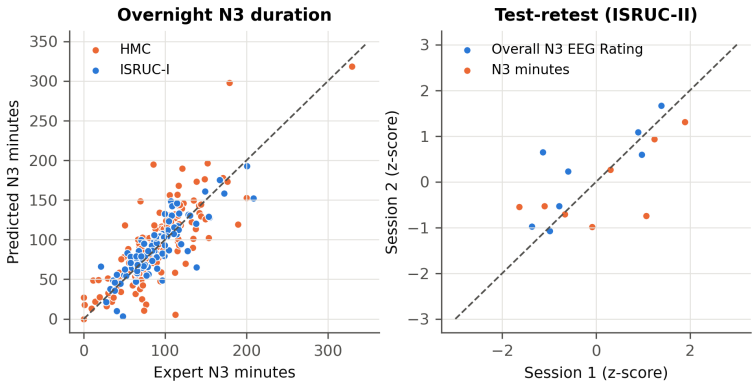


Fig. S6. Predicted vs expert N3 minutes per night (left) and ISRUC-II test–retest of the rating and of N3 minutes (right). Bland–Altman plots are in Fig. 4 of the main text.

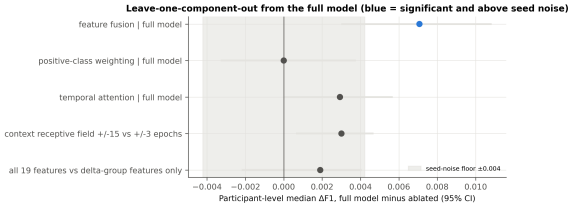


Fig. S7. **[post hoc]** Post hoc leave-one-component-out from the full model. The shaded band is the largest seed-to-seed difference of the primary model.

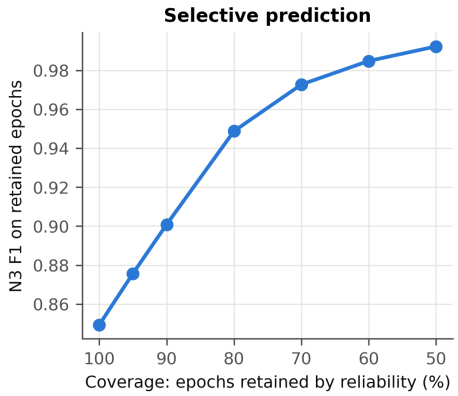


Fig. S9. N3 F1 score on the epochs retained when the least reliable epochs are withheld (cross-validation).

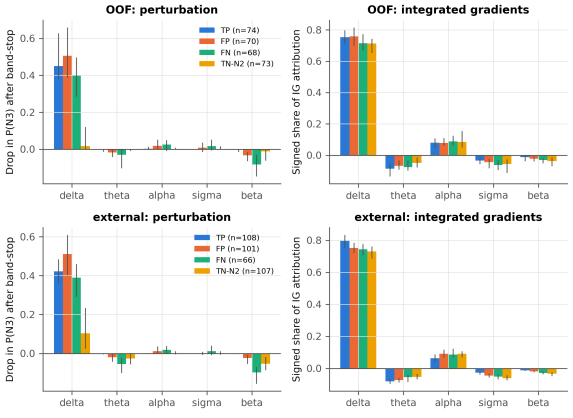


Fig. S8. **[post hoc]** Post hoc band importance by outcome (true and false positives, false negatives, true-negative N2 epochs) in cross-validation and external cohorts.

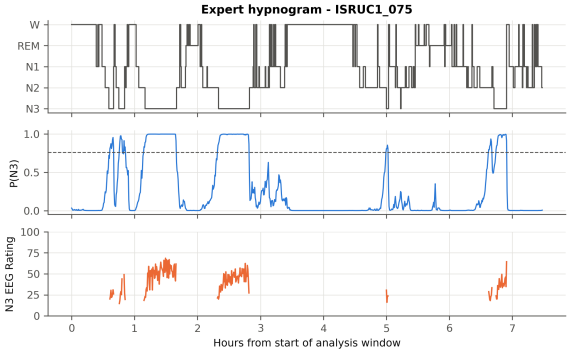


Fig. S10. One ISRUC-I night: expert hypnogram, N3 probability with the decision threshold (dashed), and the N3 EEG rating of detected epochs.